

eigenvalues of the spheroidal equation), to the Nekrasov-Shatashvili instanton counting function. More details on the connection between exact WKB and $\mathcal{N} = 2$ supersymmetric field theories is available in [57].

WKB periods can also be calculated with thermodynamic-Bethe-ansatz (TBA) equations [58–62]. The TBA approach is perhaps the most efficient approach for calculating quantum periods and solving exact quantization conditions numerically. The TBA approach however does not produce asymptotic series with full parameter dependencies included. As described in [61], the seminal work of Voros [51] provided a method for computing spectral data from exact quantization conditions using what was termed the “analytic bootstrap.” In this approach the WKB periods play a fundamental role and are Borel resummed to form exact quantization conditions. The discontinuity structure encoded by the Stokes automorphism of the WKB periods determines a Riemann-Hilbert problem which has a solution in terms of a TBA-like system [61]. The connection between TBA and differential equations, the ODE/IM correspondence [63], has been used for the calculation of quasinormal modes [37] of D3 branes and the Reissner-Nordström geometry. However, to the knowledge of the author it has not been applied to the case of Kerr black holes.

As a final comment we mention that exact quantization conditions can also be solved analytically in terms of transseries [54]. Here we lay the foundation for such computations. For this reason we focus on the computation of the quantum actions exactly and with full parameter dependence included.

2.1 Quantum actions and the Bohr-Sommerfeld condition

Applying a *WKB* ansatz $\psi(x) = e^{\frac{i}{\hbar} \int^x \phi dx}$ to the Schrödinger equation (1.1) gives the following Riccati equation

$$\phi^2 = 2(E - \cos^2 x - \mu \csc^2 x) - i\hbar \frac{d}{dx} \phi. \quad (2.1)$$

Setting $\hbar = 0$, this equation is algebraic and coincides with the classical equation of motion with the *action density* ϕ playing the role of momentum. In the complex domain the classical equation of motion defines a Riemann surface which happens to be a torus², call it Σ .

The equation (2.1) has an asymptotic series solution [50] for the action density ϕ . Let

$$\phi(\mu, E, \hbar, x) = \sum_{k=0}^{\infty} \phi_k(\mu, E, x) \hbar^k. \quad (2.2)$$

Each of the $\phi_k dx$ are differentials on the Riemann surface Σ . More precisely they are differentials on Σ where the points $x = \dots, -2\pi, -\pi, 0, \pi, \dots$ (which correspond to simple poles of the ϕ_k) are removed.³ This means that we can integrate them along paths on Σ and define quantum actions.

²The classical equation defines a two parameter family of tori.

³It is important to point out that the series has the property that the odd terms are all total derivatives with the exception of ϕ_1 .